10 In this question, the **i** direction is horizontal and the **j** direction is vertically upwards.

A particle of mass 3 kg is in equilibrium.

(a) Explain what is meant by a particle. **[1]**

The following forces are acting on the particle:

- the weight of the particle
- a horizontal force of $h\mathbf{i}$ N
- the force $\mathbf{F} = k\mathbf{i} + (k-5)\mathbf{j}$ N.

(b) Calculate the value of h. **[3]**

(c) The force **F** is removed.

State the vector equation of motion of the particle. **[1]**

- Give your final answers to a degree of accuracy that is appropriate to the context.
- The acceleration due to gravity is denoted by $g\,\text{m}\,\text{s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

10(a)

10(b)

10(c)